

Faizaan Fazal Khan

Doctoral Researcher Candidate | AI in Biomedical Imaging

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7

Journal Articles

5

Conferences

4

Intl Journals (IEEE/MDPI/Frontiers)

3

Domestic Journals (Smart Media)

4.38/4.5

M.S. CGPA

PROFESSIONAL SUMMARY

Graduate researcher specializing in AI-driven biomedical imaging with a focus on neurodegenerative disease diagnosis. Experienced in designing and optimizing 3D U-Net and EfficientNet-based deep learning pipelines for MRI and PET analysis, achieving 93.2% Dice accuracy on ADNI datasets. Author of **7 peer-reviewed journal articles** across IEEE Access, MDPI Mathematics, MDPI Biomedicines, Frontiers in Aging Neuroscience, and Smart Media Journal, with presenter credits at 5 international conferences. Proficient in PyTorch, TensorFlow, MONAI, FSL, SPM, and ANTs. Completed M.S. with 4.38/4.5 CGPA at Chosun University. Recipient of the Chosun University Academic Award 2026 for outstanding research performance. Seeking to advance explainable and clinically trustworthy AI at doctoral level.

EDUCATION

M.S. in Information and Communication Engineering

Mar 2023 – Feb 2026

Chosun University, Gwangju, South Korea | CGPA: 4.38 / 4.5

- Thesis: AI-driven diagnosis of Alzheimer's Disease using multimodal MRI and PET neuroimaging
- Recipient of the Chosun University Academic Award 2026 for top research performance in the Master's program

B.S. in Computer Science

Sep 2019 – Jun 2023

Air University Islamabad, Pakistan | CGPA: 3.22 / 4.0

RESEARCH EXPERIENCE

Graduate Research Assistant

Mar 2024 – Present

Chosun University, Dept. of Information and Communication Engineering, Gwangju, South Korea

- Designed and optimized a resource-efficient 3D U-Net for hippocampal segmentation incorporating CLAHE and SCE-3DWT preprocessing, achieving 93.2% Dice accuracy on ADNI datasets (IEEE Access, 2025).
- Conducted dataset-aware preprocessing ablation and transfer learning study across multiple ADNI cohorts; insights published in MDPI Mathematics (2025).
- Developed EfficientNetV2B0 classification pipeline combined with XAI methods (Grad-CAM, LIME) to improve clinical interpretability of Alzheimer's MRI analysis (Smart Media Journal, 2025).
- Performed longitudinal survival analysis (Kaplan-Meier, Cox regression) on 1,115 ADNI participants modeling APOE e4 allele dose effects on MCI-to-AD conversion, achieving C-index 0.805 (MDPI Biomedicines, 2026).
- Published findings on APOE e4 as a predictor of cognitive decline and hippocampal volume interaction (Frontiers in Aging Neuroscience, accepted).
- Led ablation experiments, hyperparameter tuning, and multimodal data normalization using PyTorch, TensorFlow, and MONAI on GPU servers; presented at 5 international conferences.
- Currently expanding research into explainable AI frameworks for clinical decision support and trustworthy model deployment.

PUBLICATIONS

International Journals

- [J1] Khan, F.F., Kim, J.-H., Kim, J.-I., & Kwon, G.-R. (2026). *APOE e4 Allele Dose and Time to Clinical Conversion from Mild Cognitive Impairment to Alzheimer's Disease Dementia: An ADNI Survival Analysis*. **MDPI Biomedicines**, 14(6): 1280. IF: 4.3 | doi: 10.3390/biomedicines14061280
- [J2] Khan, F.F., Kim, J.-H., Kim, J.-I., & Kwon, G.-R. *APOE e4 as a Predictor of Cognitive Decline and Its Interaction with Hippocampal Volume in Alzheimer's Disease*. **Frontiers in Aging Neuroscience**. IF: 4.1 | doi: 10.3389/fnagi.2026.1730265
- [J3] Khan, F.F., Kim, J.-H., Park, C.-S., Kim, J.-I., & Kwon, G.-R. (2025). *A Resource-Efficient 3D U-Net for Hippocampus Segmentation Using CLAHE and SCE-3DWT Techniques*. **IEEE Access**. IF: 3.4 | doi: 10.1109/ACCESS.2025.3577266
- [J4] Khan, F.F., Kim, J.-H., Kim, J.-I., & Kwon, G.-R. (2025). *Dataset-Aware Preprocessing for Hippocampal Segmentation: Insights from Ablation and Transfer Learning*. **MDPI Mathematics**, 13(20): 3309. IF: 2.3 | doi: https://doi.org/10.3390/math13203309

Domestic Journals (South Korea)

- [J5] Khan, F.F., & Kwon, G.-R. (2025). *Toward Clinically Trustworthy Alzheimer's Diagnosis: Combining EfficientNetV2B0 and XAI Techniques for MRI Analysis*. **Smart Media Journal**, 14(6): 60-66.

[J6] Khan, F.F., & Kwon, G.-R. (2024). *Comparison and Analysis of CNN Models to Address Skewed Data Issues in Alzheimer's Diagnosis*. **Smart Media Journal**, 13(10): 28-34.

[J7] Khan, F.F., & Kwon, G.-R. *Minimal Clinical Benchmark for Alzheimer's Disease Prediction Using Age and MMSE*. **Smart Media Journal**. (In press)

Conference Presentations (First Author)

- [C1] Khan, F.F., et al. KNGC 2024 (Korea Next-Generation Computing Conference), 2024.
- [C2] Khan, F.F., et al. ICNGC 2024, 2024.
- [C3] Khan, F.F., et al. ICICPE 2024, 2024.
- [C4] Khan, F.F., et al. ICSEC 2025, November 2025.
- [C5] Khan, F.F., et al. ICNGC 2025, 2025.

TECHNICAL SKILLS

Programming	Python, MATLAB, JavaScript, TypeScript, C++, C#, Java
Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, FastAI, Lightning, MONAI Transforms, Hugging Face Transformers
Medical Imaging	FSL, SPM, ANTs, Nibabel, NiftyReg, SimpleITK, 3D Slicer, ITK-Snap, MONAI
Computer Vision	OpenCV, Open3D, Image Segmentation, Detection, Classification, XAI (Grad-CAM, LIME, SHAP)
Data and Stats	Pandas, NumPy, Matplotlib, Seaborn, Survival Analysis (Kaplan-Meier, Cox Regression), Statistical Validation
Deployment	Docker, Flask, FastAPI, Streamlit
HPC and Dev Tools	GPU Server Setup, Git, GitHub, JupyterLab, Google Colab, LaTeX, Overleaf, VS Code

OTHER PROFESSIONAL EXPERIENCE

Lecturer, Computer Science	2023 – 2024
NCLS and C NUST, Islamabad, Pakistan	
• Taught programming, algorithms, and foundational AI concepts to high school students; supervised ICT-based assignments and managed academic records.	
Software Team Lead	2023 – 2026
Jazba Innovations, Remote	
• Built and deployed 10+ client websites using Next.js, React, and Node.js; improved checkout reliability by 30% and organic traffic by 45% through Shopify API and SEO optimization.	
• Led a 4-member development team with Agile sprints and Git-based version control.	

LANGUAGE PROFICIENCY

English	IELTS Academic: Overall Band 7.0 (CEFR C1) Listening 7.0 Reading 7.5 Writing 6.5 Speaking 7.5	25 Oct 2025
Urdu	Native proficiency	

AWARDS AND CERTIFICATIONS

Academic Award for Top Research Activities in Master's Program	Chosun University, 2026
Meta Advanced React Course Professional Certificate (Coursera)	2023
Coursera: Introduction to Software Engineering	2023
Coursera: Getting Started with Git and GitHub	2023
Coursera: Developing Cloud Native Applications	2023
HackerRank SQL Advanced Certificate	2022
HackerRank Python Basic Certificate	2022
Top Performer, Games of Codes, Air University Computing and Innovation Society	2022
Volunteer, Human Care Foundation	2021 – 2022